Repeated Integration by Parts: Complete Tutorial with Tabular Method

Calculus Tutorial

Contents

1 Introduction to Repeated Integration by Parts

1.1 What is Repeated Integration by Parts?

Repeated integration by parts is when you need to apply the integration by parts formula **more than once** to completely solve an integral.

This typically happens when integrating:

- Polynomials multiplied by exponentials: $x^n e^{ax}$
- Polynomials multiplied by trigonometric functions: $x^n \sin(ax)$, $x^n \cos(ax)$
- Logarithmic functions raised to powers: $(\ln x)^n$

1.2 When Do You Need Repeated Integration by Parts?

You know you need repeated integration by parts when:

1. After one application of integration by parts, you get another integral that **still requires integration by parts**.
2. The polynomial degree in the integral doesn't reduce to a constant after one application.
3. You're integrating products where one part gets simpler with differentiation (like polynomials) and the other part cycles (like trigonometric functions).

2 The Basic Strategy

2.1 Choosing u and dv (The LIATE Rule)

For repeated integration by parts, the choice of u is even more critical. Always follow LIATE:

Priority	Function Type
1	**L**ogarithmic: $\ln x$, $\log x$
2	**I**nverse Trigonometric: $\tan^{-1} x$, $\sin^{-1} x$
3	**A**lgebraic (Polynomials): x, x^2, x^3
4	**T**rigonometric: $\sin x$, $\cos x$
5	**E**xponential: e^x, a^x

2.2 The Repeated Process

For $\int x^n f(x)dx$ where $f(x)$ is exponential or trigonometric:

1. Set $u = x^n$ and $dv = f(x)dx$
2. Apply integration by parts: $\int udv = uv - \int vdu$
3. The new integral will have x^{n-1} times some function
4. Repeat until the polynomial part becomes a constant
5. Collect all terms

3 Step-by-Step Examples

3.1 Example 1: Polynomial $\times$ Exponential

$$\int x^2 e^x \, dx$$

Solution:

First application:

$$\begin{aligned}
\text{Let } u = x^2, &\quad dv = e^x dx \\
du = 2x dx, &\quad v = e^x
\end{aligned}$$

$$\begin{aligned}
\int x^2 e^x dx &= x^2 e^x - \int 2x e^x dx \\
&= x^2 e^x - 2\int x e^x dx
\end{aligned}$$

Second application (for $\int xe^x dx$):

$$\begin{aligned}
\text{Let } u = x, &\quad dv = e^x dx \\
du = dx, &\quad v = e^x
\end{aligned}$$

$$\begin{aligned}
\int x e^x dx &= x e^x - \int e^x dx \\
&= x e^x - e^x
\end{aligned}$$

Combine results:

$$\begin{aligned}
\int x^2 e^x dx &= x^2 e^x - 2(x e^x - e^x) + C \\
&= x^2 e^x - 2x e^x + 2e^x + C \\
&= e^x(x^2 - 2x + 2) + C
\end{aligned}$$

3.2 Example 2: Polynomial × Sine

$$\int x^2 \sin x \, dx$$

Solution:
First application:

$$\begin{aligned}
\text{Let } u = x^2, &\quad dv = \sin x dx \\
du = 2x dx, &\quad v = -\cos x
\end{aligned}$$

$$\begin{aligned}
\int x^2 \sin x dx &= -x^2 \cos x + \int 2x \cos x dx \\
&= -x^2 \cos x + 2\int x \cos x dx
\end{aligned}$$

Second application (for $\int x \cos x dx$):

$$\begin{aligned}
\text{Let } u = x, &\quad dv = \cos x dx \\
du = dx, &\quad v = \sin x
\end{aligned}$$

$$\begin{aligned}
\int x \cos x dx &= x \sin x - \int \sin x dx \\
&= x \sin x + \cos x
\end{aligned}$$

Combine results:

$$\begin{aligned}
\int x^2 \sin x dx &= -x^2 \cos x + 2(x \sin x + \cos x) + C \\
&= -x^2 \cos x + 2x \sin x + 2\cos x + C
\end{aligned}$$

4 The Tabular Method (DI Method)

The tabular method is a **much faster** way to handle repeated integration by parts, especially for exams.

4.1 How the Tabular Method Works

1. Create two columns: **Differentiate** and **Integrate**
2. In the Differentiate column, start with the function you would choose as u
3. In the Integrate column, start with the function you would choose as dv
4. Differentiate down the first column until you get 0
5. Integrate down the second column the same number of times
6. Multiply diagonally with alternating signs: $+, -, +, -, \ldots$

4.2 Example: $\int x^3 e^x dx$ using Tabular Method

Differentiate	Integrate
x^3	e^x
$3x^2$	e^x
$6x$	e^x
6	e^x
0	e^x

Alternating signs: $+, -, +, -, +$

$$\begin{aligned}\int x^3 e^x dx &= (+)(x^3)(e^x) + (-)(3x^2)(e^x) + (+)(6x)(e^x) + (-)(6)(e^x) \\ &= x^3 e^x - 3x^2 e^x + 6xe^x - 6e^x + C \\ &= e^x(x^3 - 3x^2 + 6x - 6) + C\end{aligned}$$

4.3 Example: $\int x^2 \cos x dx$ using Tabular Method

Differentiate	Integrate
x^2	$\cos x$
$2x$	$\sin x$
2	$-\cos x$
0	$-\sin x$

Alternating signs: $+, -, +, -$

$$\begin{aligned}\int x^2 \cos x dx &= (+)(x^2)(\sin x) + (-)(2x)(-\cos x) + (+)(2)(-\sin x) \\ &= x^2 \sin x + 2x \cos x - 2 \sin x + C\end{aligned}$$

5 Special Cases and Patterns

5.1 When Functions Cycle

For trigonometric functions like $\sin x$ and $\cos x$, the integrals cycle. After two integrations by parts, you might get back the original integral.

Example: $\int e^x \sin x dx$

After two integrations by parts:

$$\begin{aligned}
\text{Let } I &= \int e^x \sin x dx \\
\text{First: } I &= -e^x \cos x + \int e^x \cos x dx \\
\text{Second: } \int e^x \cos x dx &= e^x \sin x - \int e^x \sin x dx \\
\text{So: } I &= -e^x \cos x + e^x \sin x - I \\
2I &= e^x(\sin x - \cos x) \\
I &= \frac{e^x}{2}(\sin x - \cos x) + C
\end{aligned}$$

5.2 When to Stop in Tabular Method

- For polynomials × exponentials/trig: Stop when the Differentiate column reaches 0
- For functions that cycle (like $\sin x$, $\cos x$): Stop when you see the original integral pattern
- For logarithmic functions: The derivative eventually becomes 0

6 15 Practice Problems with Solutions

6.1 Easy Level

1. $\int xe^x dx$

   Solution:

$$\begin{aligned}
u &= x, \quad dv = e^x dx \\
du &= dx, \quad v = e^x \\
\int xe^x dx &= xe^x - \int e^x dx \\
&= xe^x - e^x + C \\
&= e^x(x-1) + C
\end{aligned}$$

2. $\int x \sin x dx$

   Solution:

$$\begin{aligned}
u &= x, \quad dv = \sin x dx \\
du &= dx, \quad v = -\cos x \\
\int x \sin x dx &= -x \cos x + \int \cos x dx \\
&= -x \cos x + \sin x + C
\end{aligned}$$

3. $\int x \cos x dx$

Solution:

$$u = x, \quad dv = \cos x dx$$
$$du = dx, \quad v = \sin x$$
$$\int x \cos x dx = x \sin x - \int \sin x dx$$
$$= x \sin x + \cos x + C$$

6.2 Medium Level

4. $\int x^2 e^x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^2	e^x
$2x$	e^x
2	e^x
0	e^x

$$\int x^2 e^x dx = x^2 e^x - 2xe^x + 2e^x + C$$
$$= e^x(x^2 - 2x + 2) + C$$

5. $\int x^2 \sin x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^2	$\sin x$
$2x$	$-\cos x$
2	$-\sin x$
0	$\cos x$

$$\int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x + C$$

6. $\int x^2 \cos x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^2	$\cos x$
$2x$	$\sin x$
2	$-\cos x$
0	$-\sin x$

$$\int x^2 \cos x dx = x^2 \sin x + 2x \cos x - 2 \sin x + C$$

6.3 Harder Problems

7. $\int x^3 e^x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^3	e^x
$3x^2$	e^x
$6x$	e^x
6	e^x
0	e^x

$$\begin{aligned}\int x^3 e^x dx &= x^3e^x - 3x^2e^x + 6xe^x - 6e^x + C \\ &= e^x(x^3 - 3x^2 + 6x - 6) + C\end{aligned}$$

8. $\int x^3 \sin x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^3	$\sin x$
$3x^2$	$-\cos x$
$6x$	$-\sin x$
6	$\cos x$
0	$\sin x$

$$\int x^3 \sin x dx = -x^3\cos x + 3x^2 \sin x + 6x\cos x - 6\sin x + C$$

9. $\int x^3 \cos x dx$

Solution:

Tabular method:

Differentiate	Integrate
x^3	$\cos x$
$3x^2$	$\sin x$
$6x$	$-\cos x$
6	$-\sin x$
0	$\cos x$

$$\int x^3 \cos x dx = x^3 \sin x + 3x^2\cos x - 6x\sin x - 6\cos x + C$$

6.4 Very Common Exam Types

10. $\int x^n e^{ax} dx$

General Pattern:

$$\int x^n e^{ax} dx = e^{ax}\left(\frac{x^n}{a} - \frac{nx^{n-1}}{a^2} + \frac{n(n-1)x^{n-2}}{a^3} - \cdots + (-1)^n \frac{n!}{a^{n+1}}\right) + C$$

11. $\int e^x \sin x dx$

Solution:

$$\begin{aligned}
\text{Let } I &= \int e^x \sin x dx \\
\text{First: } u &= e^x, \quad dv = \sin x dx \\
du &= e^x dx, \quad v = -\cos x \\
I &= -e^x \cos x + \int e^x \cos x dx \\
\text{Second: } u &= e^x, \quad dv = \cos x dx \\
du &= e^x dx, \quad v = \sin x \\
\int e^x \cos x dx &= e^x \sin x - \int e^x \sin x dx \\
&= e^x \sin x - I \\
\text{So: } I &= -e^x \cos x + e^x \sin x - I \\
2I &= e^x(\sin x - \cos x) \\
I &= \frac{e^x}{2}(\sin x - \cos x) + C
\end{aligned}$$

12. $\int e^x \cos x dx$

Solution:

$$= \frac{e^x}{2}(\sin x + \cos x) + C$$

13. $\int xe^{2x} dx$

Solution:

$$\begin{aligned}
u &= x, \quad dv = e^{2x} dx \\
du &= dx, \quad v = \frac{1}{2}e^{2x} \\
\int xe^{2x} dx &= \frac{x}{2}e^{2x} - \frac{1}{2}\int e^{2x} dx \\
&= \frac{x}{2}e^{2x} - \frac{1}{4}e^{2x} + C \\
&= \frac{e^{2x}}{4}(2x - 1) + C
\end{aligned}$$

14. $\int x^2 e^{-x} dx$

Solution:

Tabular method:

Differentiate	Integrate
x^2	e^{-x}
$2x$	$-e^{-x}$
2	e^{-x}
0	$-e^{-x}$

$$\begin{aligned}
\int x^2 e^{-x} dx &= -x^2 e^{-x} - 2xe^{-x} - 2e^{-x} + C \\
&= -e^{-x}(x^2 + 2x + 2) + C
\end{aligned}$$

15. $\int (\ln x)^2 dx$

Solution:

$$\begin{aligned} u = (\ln x)^2, &\quad dv = dx \\ du = \frac{2\ln x}{x}dx, &\quad v = x \end{aligned}$$

$$\begin{aligned} \int (\ln x)^2 dx &= x(\ln x)^2 - 2\int \ln x dx \\ &= x(\ln x)^2 - 2(x\ln x - x) + C \\ &= x(\ln x)^2 - 2x\ln x + 2x + C \end{aligned}$$

7 Exam Tips and Tricks

7.1 Quick Recognition Guide

Integral Type	**Method**
$x^n e^{ax}$	Tabular method (n+1 rows)
$x^n \sin(ax)$ or $x^n \cos(ax)$	Tabular method (n+1 rows)
$e^{ax}\sin(bx)$ or $e^{ax}\cos(bx)$	Integration by parts twice, then solve for I
$(\ln x)^n$	Integration by parts, u = $(\ln x)^n$
$x^n \ln x$	Integration by parts, u = $\ln x$

7.2 One-Line Exam Rule

Polynomial $\times$ sin/cos/exp $\rightarrow$ Use Tabular Method for Repeated Integration by Parts

7.3 Common Mistakes to Avoid

1. **Not choosing u correctly:** Always use LIATE rule
2. **Forgetting the alternating signs:** In tabular method, signs alternate: +, , +, , ...
3. **Not simplifying:** Always factor common terms (like e^x)
4. **Stopping too early:** In tabular method, differentiate until you get 0
5. **Forgetting the constant of integration:** Always add +C

8 Practice for Mastery

To master repeated integration by parts:

1. Practice the tabular method until it becomes automatic
2. Memorize common patterns for x^2e^x, $x^2\sin x$, etc.
3. Always check your answer by differentiating
4. Time yourself to build exam speed